



Islamic University Of Technology

EEE 4308: Digital Electronics

Project Name: SAP – 1

Team: Digital Electronauts

Phase - A

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Phase - A

Objective:

- 1) Learning about Clock of SAP-1
- 2) Learning about Program counter of SAP-1
- 3) Gaining knowledge about D-latch
- 4) Gaining knowledge about D Flip-flop
- 5) Working of Clock & Counter inside the SAP

Description:

This project is about understanding of 6 bit computer and construction of it. Basically its about understanding inside process of a computer. We can understand what does it mean clock speed of a computer and how we can control this speed. For running this computer we need to program the RAM first, then we can run the computer. Program is stored inside RAM as a set of instruction. Other modules acts according to instruction. Program counter counts the No. of instruction and MAR sends the memory address for next instruction. Instructed operation is occurred inside ALU. Value of ALU is stored inside Accumulator. When necessary, Accumulator sends the data to output register to show the result. Necessary signal for all the module is sent by controller sequencer. Controller sequencer generates signals according to instruction code sent by instruction register. All the module is runed by clock pulse. Without clock pulse this computer can't work. There is HLT operation in our computer that stopes the main clock, as a result whole computer stopes. This is total working system of our SAP-1.

1)Clock Circuit:

This clock circuit generates the main clock signal for the computer. NE555 timer is used to built this circuit. It has two modes: 1) Auto and 2) Manual . Mainly there is two type of clk signal is generated, one is astable clk and another one is monostable clk. This generated clk goes to all other modules. There is a HLT signal which stops this clk as a result whole computer stopes.

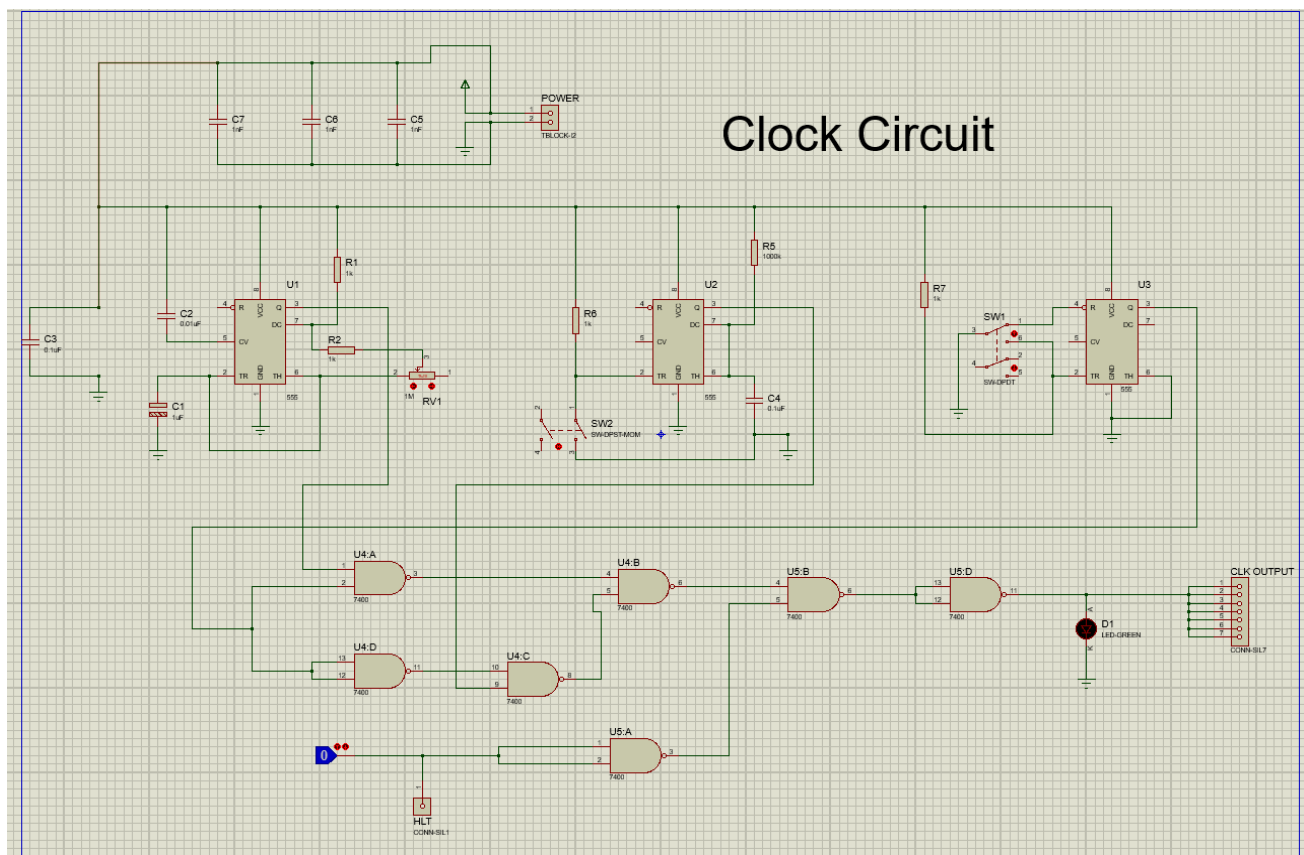
Control Bits:

No control bits.

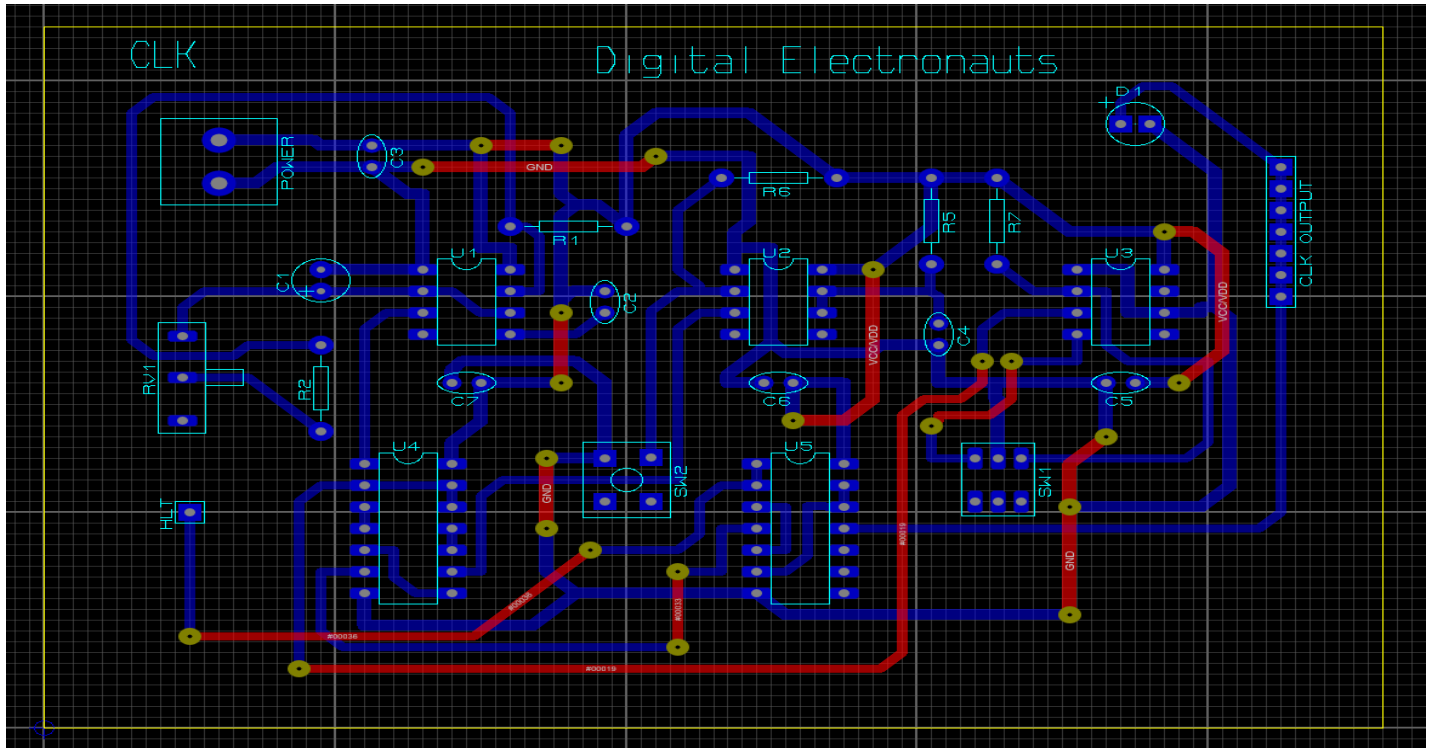
Circuit:

To built this we are using three 555 timer IC, 7400 (NAND gate), one LED and two switches.

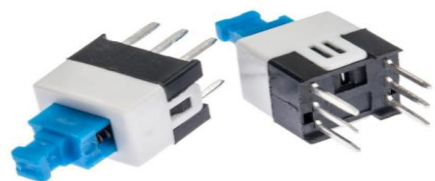
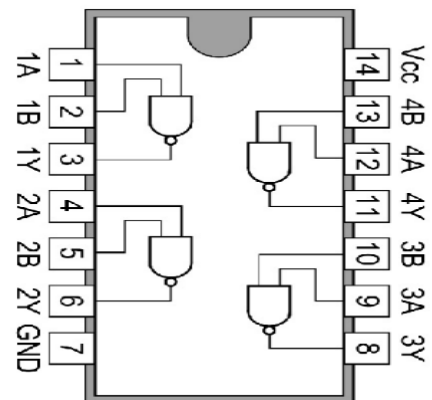
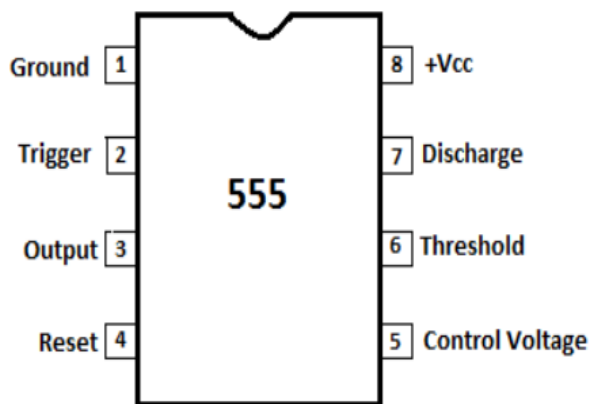
Counter Circuit in Proteous:



PCB :



Used IC's :



2)Counter:

It is a 3-bit counter circuit which can count from 000 to 111. This number is sent to MAR, MAR sent this to RAM for next instruction.

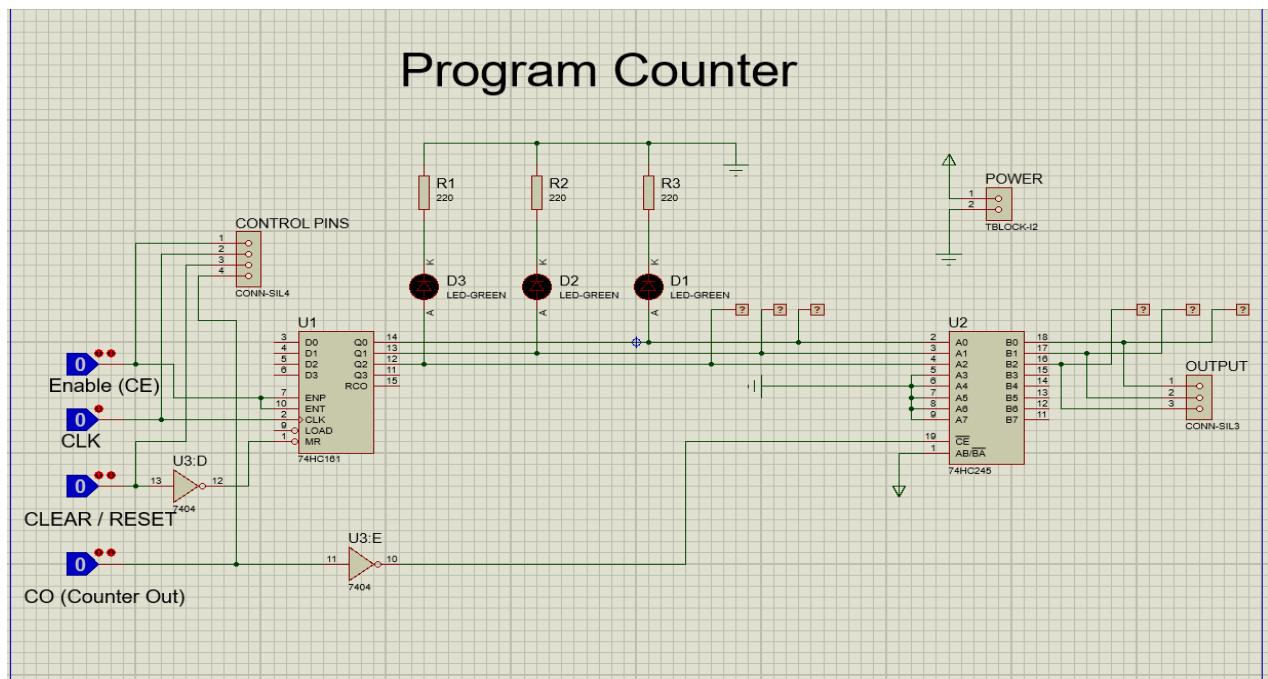
Control Bits:

- 1) **CE**: Increments its value on next clock cycle
- 2) **CO**: Sends its current stored value to BUS

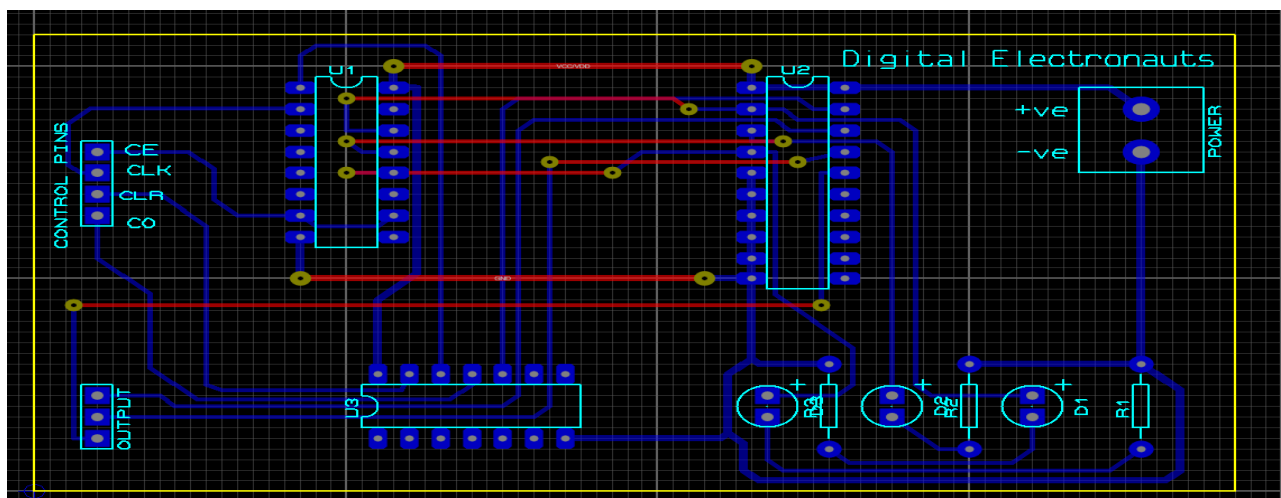
Circuit:

To built this we are using one 74HC161 IC, one 74LS245 and one 7404 (NOT Gate). 74245 is Octal BUS transceiver (Tri-state Buffer).

Counter Circuit in Proteous:

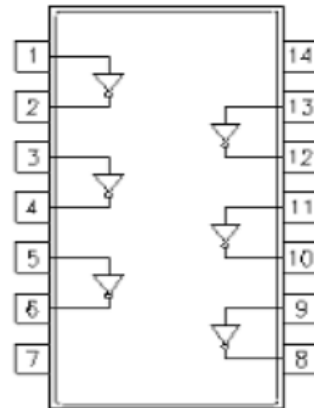
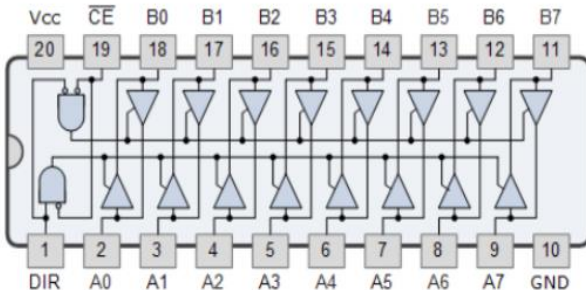


PCB :



Used IC's :

74LS245 Octal Tri-state Buffer



7404

3) MAR:

It is a 3-bit register which sends the address of next instruction to RAM.

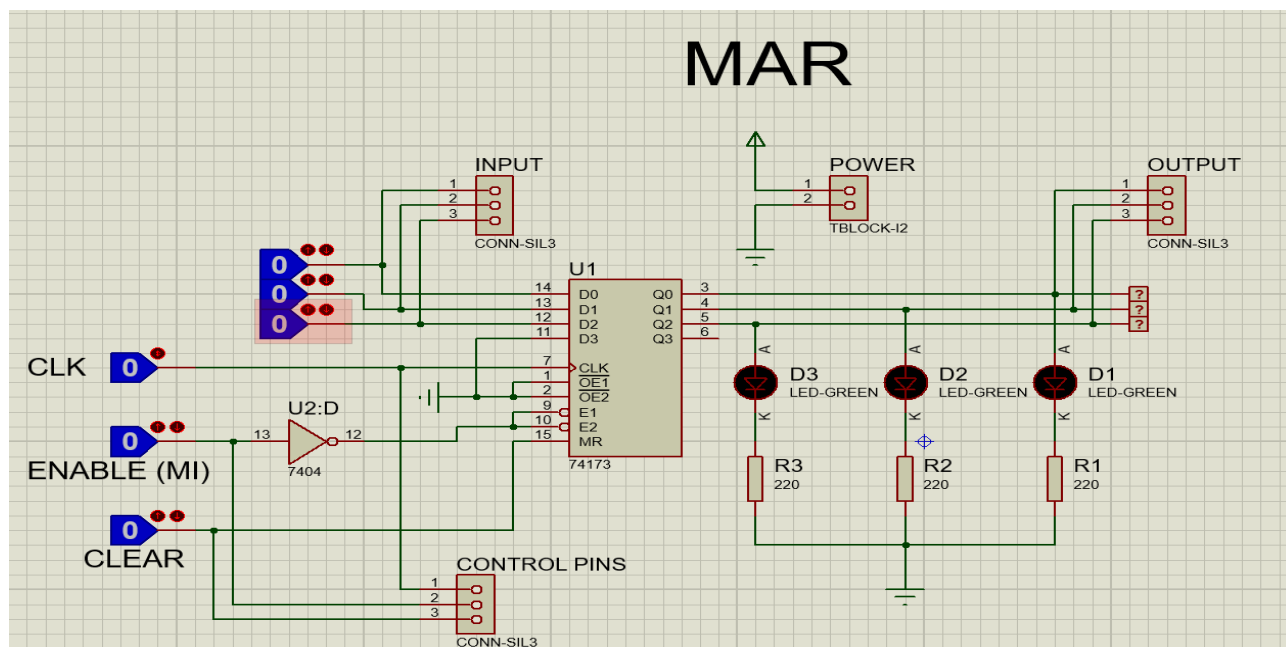
Control Bits:

- 1) **MI:** Stores current Data from BUS

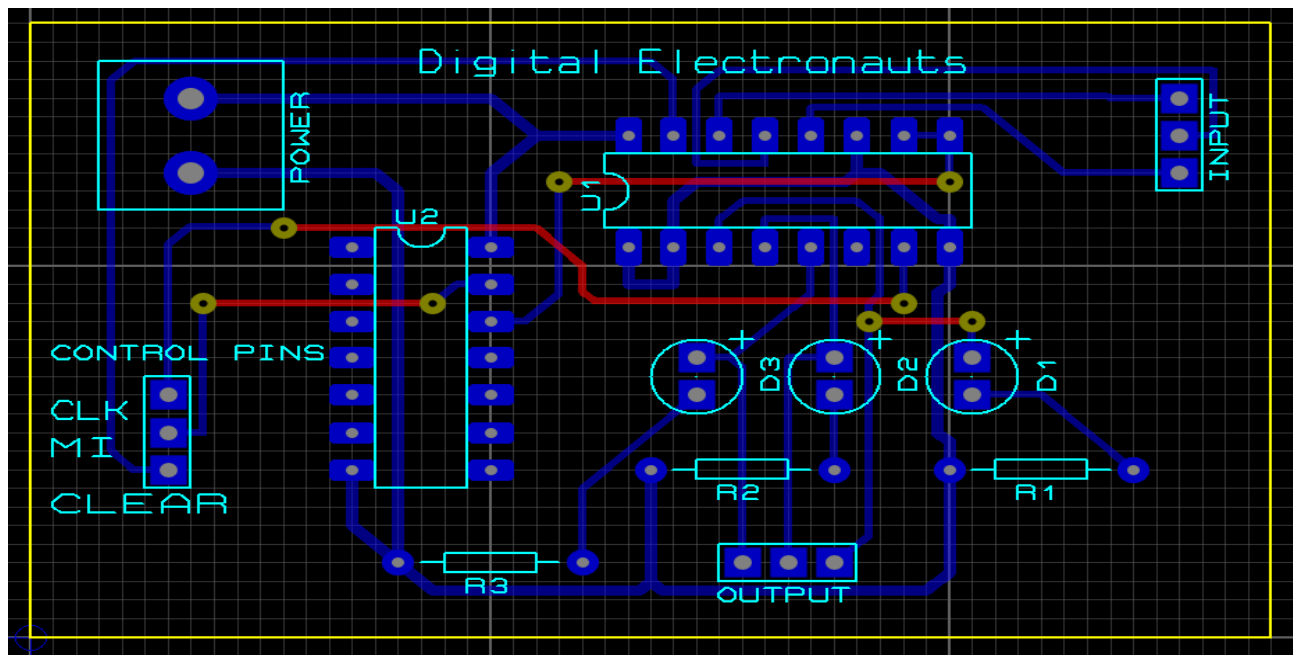
Circuit:

To built MAR we are using 74173 IC and one 7404 (NOT Gate) and Three LED & Resistors. 74173 is a 4-bit register IC.

Circuit in Proteous:



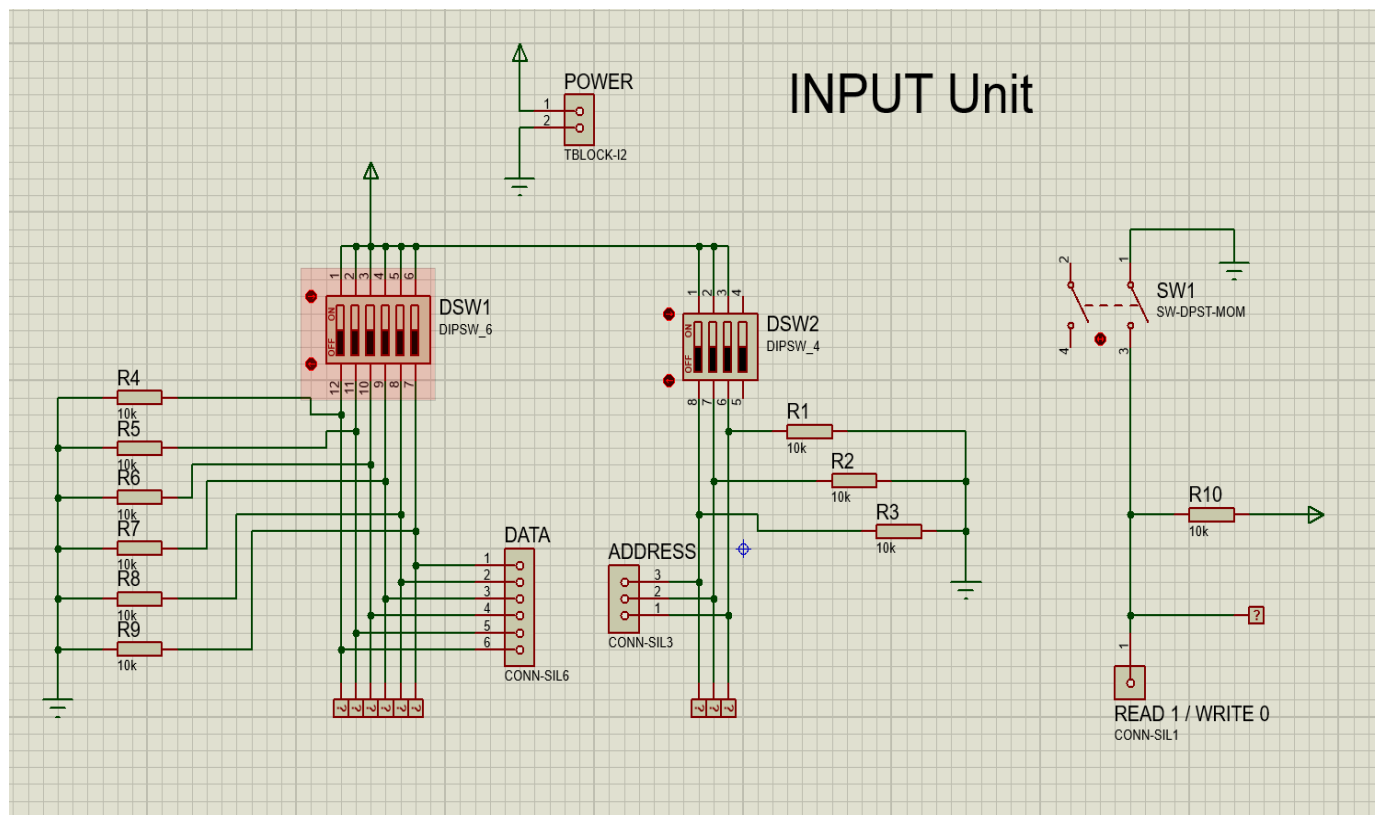
PCB:



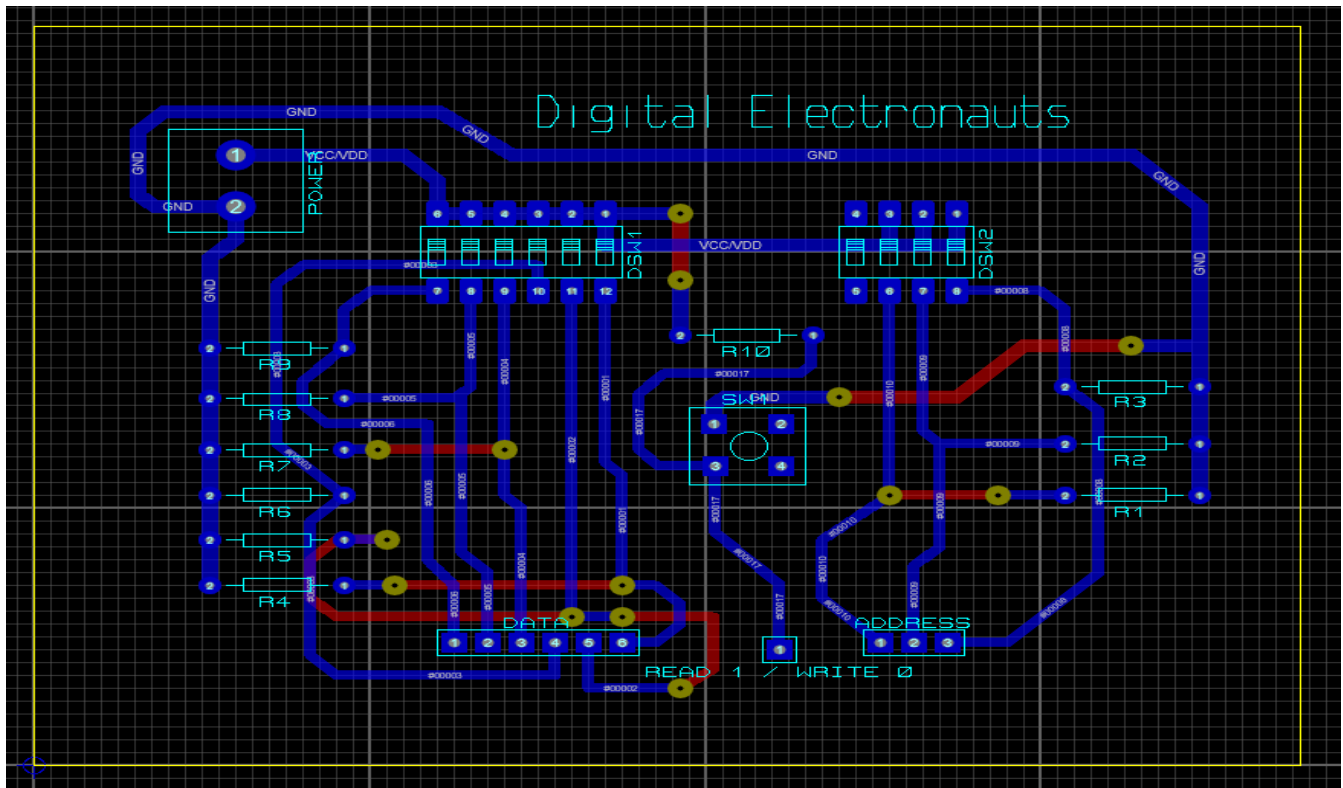
4) Input Unit:

It is used to write program in the RAM

Circuit in Proteous:



PCB:



Problem faced:

- 1) Selecting the proper IC's
- 2) Component placement
- 3) PCB design
- 4) Some components don't have PCB package
- 5) While routing the PCB

Discussion:

In this phase we designed the clock pulse generator, program counter, MAR and input unit of our SAP. Among all the modules clock pulse generator designing was little difficult one and input unit was the easiest one. It is a 6-bit SAP, we can develop it further to 8 bit, as we same number of components and concepts to built 8 bit SAP. Through this phase we learnt how a computer runs on a clock pulse. We also learnt why clock pulse is necessary for properly running a computer. It was a great learning phase for us.

Contribution of each member is added with Phase B report